SOFTWARE DESIGN & ARCHITECTURE (SESD-242)

I.Course Details

Credit Hours	3 (2+1)
Pre-requisite(s)	Software Design & Architecture
Course Leader	Dr. Huma Hayyat Khan
Recommended Book(s)	Software Engineering design theory and practices by Carles e Otero
Reference Books	1. Applying UML And Patterns: An Introduction to Object Oriented Analysis And Design And Iterative Development, Craig Larman, 3rd Ed, Pearson Education, 2005. 2. Documenting Software Architectures: Views and Beyond By Paul Clements, Felix Bachmann, Len Bass, 2nd Edition, 2002 3. Design Patterns: Elements of Reusable Object-Oriented Software, Erich Gamma, John Vlissides, Richard Helm, Ralph Johnson, 1994

II.Course Learning Outcomes (CLOs)

CLOs	Description	Domain	Taxonomy Level	PLOs	Assessment Artifacts
CLO-1	**Understand** fundamental design and architecture concepts in software systems	Cognitive	2	1	A1, Q1, Midterm exam
CLO-2	**Apply** UML to Comprehend the design challenges as well as document the design of software systems	Cognitive	4	5	A2, Q2, Midterm exam
CLO-3	**Analyze** software design patterns and architectural styles for particular and complex software design problems	Cognitive	3	2	A3, Q3, final term

III.Course Assessment

Evaluation Methods	Weight (%)
Quizzes	10
Assignments	10
Presentation/Project	5
Midterm	25
Final Term	50
Total	**100**

IV. Grading Policy

For students admitted in Fall 2021 and onwards

Grade	A+	A	B+	B	C+	C	D+	D	F
%age	>=90	80-89	75-79	70-74	65-69	60-64	55-59	50-54	<50
GPA	4.00	4.00	3.50-3.99	3.00-3.49	2.50-2.99	2.00-2.49	1.50-1.99	1.00-1.49	0.00

For students admitted before Fall 2021

Grade	A1	A2	A3	B1	B2	B3	C1	C2	D	F
%age	>=90	80-89	77-79	74-76	70-73	67-69	64-66	60-63	50-59	<50
GPA	4.00	4.00	3.66	3.33	3.00	2.66	2.33	2.00	1.50	0.00

V. Course Contents

Software Design and Architecture Concepts, Design principles, Object-Oriented Design with UML, Architecture views, User interface design, Persistent layer design, Web applications design, State machine diagrams and modeling, Design Patterns, Architectural design issues, , Software Architecture, Architectural Structures & Styles-, Architectural Patterns.

VI. Weekly Breakdown

Week No.	CLOs	Topics	Reference
1	CLO-1	Introduction to Software Engineering Design, Conceptual Overview, engineering Design, Engineering Problem Solving, Initial State, Operational State, thinking about the Problem, Problem Solution, goal State	Chapter 1
2		Introduction to Software design with UML, Conceptual Overview, what Is UML? Why Study UML? The UML's Fundamentals, Classification of UML diagrams. Designing the Architecture, the 4 + 1 View Model, user View, process View., physical View, development View, Logical View,	Chapter 2
3	CLO-2	Introduction to use case Identification, Use case Writing (fully dressed use case)	Chapter 6 [Ref. Book 1]
4		Introduction to use case modelling. Use case model generation	Chapter 6 [Ref. Book 1]
5		Introduction to Domain Model. Domain model generation	Chapter 6 [Ref. Book 1]
6		Introduction to class diagrams (logical view),Linking classes of Class Diagram through associations , class diagram generation	Chapter 16 [Ref. Book 1]

7		Introduction to Activity Diagram (process view), Fork and join Swim Lanes, Activity Diagram Generation	Chapter 28 [Ref. Book 1]
8		Introduction to Sequence diagram (process view), Message passing and its types, Loop and condition in sequence diagram, frames in sequence diagram Practice Problems	Chapter 10 [Ref. Book 1]
9	**Mid Term Exams**		
10	CLO-2	Introduction to state diagram (process view), Naming conventions Practice problems	Chapter 13 Chapter 29 [Ref. Book 1]
11		Introduction to Development View(Component Diagram) using UML	Chapter 37 [Ref. Book 2]
12		Introduction to Physical View(Deployment Diagram) using UML	Chapter 37 [Ref. Book 2]
13	CLO-3	Patterns and Styles in Software Architecture Conceptual Overview, Architectural and Design Patterns. Layered Pattern, Client Server, master slave, pipe filter, peer to peer, model view controller.	Chapter 4
14		Human Computer Interface Design; Conceptual Overview, What Is Human–Computer Interface Design?, Why Study Human–Computer Interface Design?, General HCI Design Principles. HCI design generation	Chapter 9
15		HCI design Continuation.	Chapter 9
16		Presentations Week	